BMP180 Data Sheet + Learning Requirements:

Table 1: Operating conditions, output signal and mechanical characteristics

Parameter	Symbol	Condition	Min	Typ	Max	Units
Operating temperature	T_A	operational	-40		+85	°C
		full accuracy	0		+65	
Supply voltage	V_{DD}	ripple max. 50mVpp	1.8	2.5	3.6	V
	V_{DDIO}		1.62	2.5	3.6	V
Supply current @ 1 sample / sec. 25°C	I_{DDLOW}	ultra low power mode		3		µA
	I_{DDSTD}	standard mode		5		µA
	I_{DDHR}	high resolution mode		7		µA
	I_{DDUHR}	Ultra high res. mode		12		µA
	I_{DDAR}	Advanced res. mode		32		µA
Peak current	I_{peak}	during conversion		650	1000	µA
Standby current	I_{DDSBM}	@ 25°C		0.1		µA
Relative accuracy pressure V_{DD} = 3.3V		950 . . . 1050 hPa @ 25 °C		±0.12		hPa
				±1.0		m
		700 ... 900hPa 25 . . . 40 °C		±0.12		hPa
				±1.0		m
Absolute accuracy pressure V_{DD} = 3.3V		300 . . . 1100 hPa 0 . . . +65 °C	-4.0	-1.0*	+2.0	hPa
		300 . . . 1100 hPa -20 . . . 0 °C	-6.0	-1.0*	+4.5	hPa
Resolution of output data		pressure		0.01		hPa
		temperature		0.1		°C
Noise in pressure		see table on page 12-13				
Absolute accuracy temperature V_{DD} = 3.3V		@ 25 °C	-1.5	±0.5	+1.5	°C
		0 . . . +65 °C	-2.0	±1.0	+2.0	°C
Conversion time	$t_{c_p_low}$	ultra low power mode		3	4.5	ms
pressure	$t_{c_p_std}$	standard mode		5	7.5	ms
	$t_{c_p_hr}$	high resolution mode		9	13.5	ms
	$t_{c_p_luhr}$	ultra high res. mode		17	25.5	ms
	$t_{c_p_ar}$	Advanced res. mode		51	76.5	ms
Conversion time temperature	t_{C_temp}	standard mode		3	4.5	ms
Serial data clock	f_{SCL}				3.4	MHz
Solder drifts		Minimum solder height 50µm	-0.5		+2	hPa
Long term stability**		12 months		±1.0		hPa

Supply current: how much power the system needs while it is continuously running (steady-state current)

Standby current: an important factor in battery life if the sensor spends most of its time idle. Our pressure drop sensor will spend 99.9% of its time idle. (idle-state)

- Determines how long the system can survive between charges

Relative accuracy: the sensor's precision
Absolute accuracy: difference between the value the sensor measures and Earth's absolute atmospheric pressure

Conversion time pressure: how long it takes the sensor to sample and calculate a measurement

Temperature: between -40 and 85 degrees C

Noise table for BMP150:

Table 3: Overview of BMP180 hardware accuracy modes, selected by driver software via the variable *oversampling_setting*

Mode	Parameter *oversampling_setting*	Internal number of samples	Conversion time pressure max. [ms]	Avg. current @ 1 sample/s typ. [µA]	RMS noise typ. [hPa]	RMS noise typ. [m]
ultra low power	0	1	4.5	3	0.06	0.5
standard	1	2	7.5	5	0.05	0.4
high resolution	2	4	13.5	7	0.04	0.3
ultra high resolution	3	8	25.5	12	0.03	0.25

Internal number of samples: how many samples the system takes internally and averages before giving me a single data point.

RMS noise: jitter in the data when the sensor is completely still (the smaller, the more stable/clean the data).

Bosch BMP390 Data Table:
https://www.digikey.com/en/products/detail/bosch-sensortec/BMP390/16164575

Table 2: General electrical parameter specifications

OPERATING CONDITIONS BMP390						
Parameter	**Symbol**	**Condition**	**Min**	**Typ**	**Max**	**Unit**
Operating temperature range	T_A	operational	-40	25	+85	°C
		full accuracy	0		+65	
Operating pressure range	P	full accuracy	300		1250	hPa
Sensor supply voltage	V_{DD}	ripple max. 50mVpp	1.65	1.8	3.6	V
Interface supply voltage	V_{DDIO}		1.2	1.8	3.6	V
Sleep current	I_{DDSL}	VDD = VDDIO = 1.8 V		1.4		µA
	I_{DDSL}	VDD = VDDIO = 3.6 V		1.5		µA
Peak current	I_{peak}	during pressure measurement		660	730	µA
Peak current	I_{DDT}	during temperature measurement		240	320	µA
Relative accuracy pressure[1]	A_{rel}	700 . . . 1100 hPa 25 . . . 40 °C, 10 kPa steps		±0.03		hPa
				+/-25 cm		cm
Offset temperature coefficient[2]	TCO	900 hPa 25 . . . 40 °C		±0.6[2]		Pa/K
Absolute accuracy	A^P_{full}	900 hPa 25 °C		±0.33		hPa

Temperature range: (same as BMP180)
Pressure range: 300 to 1250

FIFO: First-in, first-out buffer.